

**Q #1****Multiple Choice Type****Award: 1****Penalty: 0.33****Data Structures**

A Binary Search Tree is formed using the distinct keys  $\{10, 20, 30, 40, 50, 60\}$ . What is the largest possible value for the node that is the left child of the right child of the root?

A. 30

B. 40

C. 50

D. 60

**Your Answer:****Correct Answer: C****Not Attempted****Time taken: 00min 00sec****Discuss**

**Q #2**

**Multiple Choice Type**

**Award: 1**

**Penalty: 0.33**

**Data Structures**

Which of the following methods is a correct and efficient algorithm to verify if a given binary tree is a valid Binary Search Tree (BST)?

- A. For each node, check that its value is greater than its left child's value and less than its right child's value.
- B. Perform a post-order traversal of the tree and check if the resulting sequence of values is sorted.
- C. Traverse the tree and check if the value of every node in the left subtree is less than the root and the value of every node in the right subtree is greater than the root.
- D. Perform an in-order traversal of the tree and check if the resulting sequence of values is sorted in non-decreasing order.

**Q #3**

Multiple Choice Type

Award: 1

Penalty: 0.33

Data Structures

What is the number of distinct Binary Search Tree(s) that can be formed using the keys 2, 4, 6, 8, and 10 ?

- A. 6
- B. 14
- C. 42
- D. 120

Your Answer:

Correct Answer: C

Not Attempted

Time taken: 00min 00sec

[Discuss](#)**Q #4**

Multiple Choice Type

Award: 1

Penalty: 0.33

Data Structures

In-order traversing a tree resulted in E A C K F H D B G; the preorder traversal would return.

- A. F A E K C D B H G
- B. F A E K C D H G B
- C. E A F K H D C B G
- D. F E A K D C H B G

**Q #5**

Multiple Choice Type

Award: 1

Penalty: 0.33

Data Structures

Let  $C(X)$  be the number of leaves in a binary tree rooted at  $T$ . Assume that  $\text{IsLeaf}(T)$  returns 1 if  $T$  is a leaf. Which of the following observations leads to a recursive implementation?

- A.  $C(T) := C(T.\text{Left}) + C(T.\text{Right})$
- B.  $C(T) := C(T.\text{Left}) + C(T.\text{Right}) + 1$
- C.  $C(T) := C(T.\text{Left}) + C(T.\text{Right}) + \text{IsLeaf}(T)$
- D.  $C(T) := C(T.\text{Left}) + C(T.\text{Right}) + \text{IsLeaf}(T) + 1$

Your Answer:

Correct Answer: C

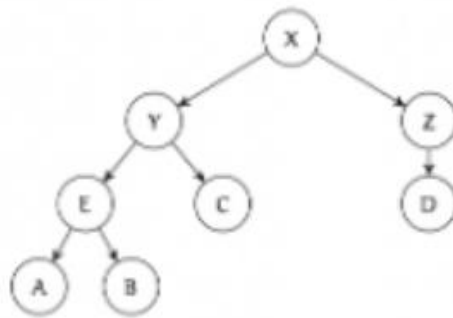
Not Attempted

Time taken: 00min 00sec

Discuss

**Q #6****Multiple Select Type****Award: 2****Penalty: 0****Data Structures**

In a level order traversal of a tree, nodes are traversed level-wise from the root and left to right in a level. For example, the level order traversal of the binary tree shown below is X, Y, Z, E, C, D, A, B.



Which one of the following combinations uniquely defines a binary tree?

- A. Level-order and pre-order traversals
- B. Level-order and in-order traversals
- C. Level-order and post-order traversals
- D. Pre-order and post-order traversals

**Q #8**

Multiple Choice Type

Award: 2

Penalty: 0.67

Data Structures

Consider the following equation, where  $N$  is the number of nodes in a particular binary search tree, and  $d(x_i)$  is the depth of node  $x_i$  in the tree. What does this equation compute?

Let All nodes in the tree are equally likely to be searched for.

$$\left( \sum_{i=1}^N d(x_i) \right) / N$$

- A. The average depth of a node in a specific BST with  $N$  nodes
- B. The average depth of a node in the average BST with  $N$  nodes
- C. The total depth of all nodes in a specific BST with  $N$  nodes
- D. The average total depth of all nodes in all BSTs with  $N$  nodes

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 00sec

Discuss

**Q #7**

Multiple Choice Type

Award: 2

Penalty: 0.67

Data Structures

We consider log to the base of 2. Thus for  $\log N$  to be a natural number,  $N$  must be a power of 2. Let  $N$  be an arbitrary number with  $N > 1$ . Which one of the following statements is true? (Assume root is at height zero)

- A. A binary tree of height  $\log N$  can have exactly  $N$  nodes.
- B. A binary tree of height  $\log N$  can have exactly  $\log N$  nodes.
- C. A binary tree of height  $\log N$  can have exactly  $N \log N$  nodes.
- D. A binary tree of height  $\log N$  can have exactly  $N^2$  nodes.

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 23sec

Discuss

**Q #9****Multiple Select Type****Award: 2****Penalty: 0****Data Structures**

Suppose we have a complete binary tree, **X**, that is not a heap, and we wish to heapify it. The obvious thing to do is just insert all nodes of **X** into a new binary heap **Y**. This works, but it doubles the space requirement. Suppose we'd prefer to heapify in place. Which of the following procedures will convert **X** into a heap (containing all of the original values)?

- A. Sink (heapify down) all nodes in level order (first the root, then its left child, then the right child of the root, etc).
- B. Swim (heapify up) all nodes in level order.
- C. Swim all nodes in reverse level order.
- D. Sink all nodes in reverse level order.

**Your Answer:****Correct Answer: B;D****Not Attempted****Time taken: 00min 00sec****Discuss****Q #10****Numerical Type****Award: 2****Penalty: 0****Data Structures**

Number of AVL tree(s) containing elements with keys 2, 4, 6, 8, and 10.

**Your Answer:****Correct Answer: 6****Not Attempted****Time taken: 00min 00sec****Discuss**